

Can My Robot Relax Me? Comparing Robotic and Human Self-Touch for Stress Reduction

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Figure 1: Overview of the study setup and our forward-looking concept. Left: experimental setup with a chair-side 6-DoF robotic arm, a 3D-printed hand end-effector with force sensing, wearable physiology sensors (BVP, EDA, respiration), and a button for user-triggered arm engagement. Middle: the robot performs a chin-support “thinking gesture,” illustrating stress-to-relief self-regulation via robotic support. Right: concept sketch of future everyday “self-affection” robotic arms enabling gesture support and comfort interactions across contexts.

Abstract

Self-touch is common and often unconscious: people hold their head when distressed, cover their face when embarrassed, or rest a hand on their chin while thinking. Prior work suggests that such body–body interactions can regulate emotions and support cognition under demanding tasks. While robotic augmentation can extend physical capability and enable agency and ownership, it remains unclear whether robot-body interactions preserve the affective benefits of body–body interactions and how users interpret these actions. We conducted an exploratory study using a stress-induction and word-recall task, in which participants performed a

head-support “thinking gesture” with either their own hand or a supernumerary robotic arm serving as a third limb. We measured task performance and multimodal physiological responses. Results provided preliminary evidence that robotic support reduces stress-related responses, yet participants consistently view the arm as an external support rather than a self-extension. This suggests robotic touch could aid emotion regulation without strong embodiment.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; **User studies**; **Accessibility**; • **Computer systems organization** → **Robotics**.

Keywords

Robotic touch, self-touch, supernumerary robotic limbs, stress regulation, embodiment, thinking gesture.

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1 Introduction

Self-touch is common and often intuitive: people rest a hand on their chin when thinking or hold their head when distressed. Prior work suggests these actions can support thinking under load and reduce stress responses [11, 14, 19]. Research on embodiment and extended cognition suggests bodily experience is flexible: with the right multisensory and sensorimotor cues, tools, and avatars can sometimes feel partially “included” in the body representation [7, 9, 16, 18, 21]. This motivates robotic augmentation, where robots are not only treated as external tools but, under strong senses of agency and ownership, can be understood as technologies that enhance the human body—for example, supernumerary or assistive arms that act as a third arm” to support everyday activities [1, 2].

However, moving from functional help (reach, manipulation) to affective support (comfort, calming) is less clear [5, 6, 15]. Work on affect-oriented robotic touch (e.g., hugging robots) shows softness, warmth, and timing can influence comfort and stress buffering [4]. If a robot makes people feel calmer, is it mainly due to material and appearance, or can a “third limb” reproduce the regulatory benefits of self-touch-like gestures? How do users perceive such interactions with a robot arm?

In this paper we define *robotic “self-touch”* as a user-triggered interaction in which a supernumerary robotic arm executes a head-support gesture, providing visual feedback (the arm moving into position in peripheral view) and tactile feedback (contact force on the face) analogous to resting one’s own hand on one’s chin. Unlike biological self-touch, the robotic touch lacks proprioceptive integration and uses a different surface material, placing it between true self-touch and externally administered touch. If robotic augmentation can reproduce the stress-buffering effects of self-touch, assistive arms could broaden their scope from “functional tools” to health-oriented interfaces that support everyday self-regulation.

To address this gap, we ran a within-subjects exploratory study around a head-support “thinking gesture” during a stress-induction and word-recall task: (A) Bare Hand, (B) No Gesture, and a supernumerary robotic arm as a supporting “third limb” with either a bare mechanical appearance (C) or a soft sleeve (D), using matched hand/trajectory/timing. Accordingly, we focus on two research questions aligned with our design:

RQ1 (Stress regulation in physiology and self-report). Compared with Bare Hand and a No Gesture control, to what extent do robotic head-support gestures modulate stress-related physiological responses and participants’ self-reported stress/workload during a high-demand task?

RQ2 (User interpretation of robotic “self-touch”). How do participants interpret and describe using a robotic arm to perform a self-touch-like head-support gesture (e.g., external support/tool vs. self-like extension), and how are these interpretations reflected

in interview themes and interaction behavior (e.g., engagement, leaning/contact force, appearance)?

We measured task performance, physiology (BVP/HRV, EDA, respiration), workload, and interviews; as a small-sample exploratory study, we focus on feasibility and preliminary patterns. Our results suggest that robotic head support may attenuate stress-related responses, but participants described the arm as an external aid rather than a self-extension, indicating potential emotional regulation benefits even without strong embodiment.

2 Related Work

2.1 Embodiment and Self–Other Boundary in Human–Robot Interaction

Prior work shows that cognitive and bodily boundaries are plastic. The “extended mind” theory suggests artifacts can function as parts of the cognitive system [9], while the rubber hand illusion shows multisensory synchrony can induce ownership and proprioceptive drift [7]. Tool use expands peripersonal space [18], and VR paradigms quantify ownership and agency [16]. However, it remains unclear whether artificial extensions can match biological limbs in affective domains, specifically stress and emotion regulation through touch, despite known self–other mechanisms such as predictive attenuation [3, 26].

2.2 Self-Touch and Thinking Gestures for Stress Regulation and Cognition

Behavioral and neural science research findings indicate that gestures and self-touch serve as cognitive scaffolds and emotional regulators. For instance, co-speech gestures reduce working memory load during problem-solving [11, 22], while spontaneous facial self-touch increases with cognitive demand, correlating with transient neural state shifts [19]. Affectively, self-soothing touch and hugs have been shown to lower cortisol levels and attenuate threat-related brain responses [14].

This raises a critical question within the extended mind framework: can a robotic limb replicate these biological regulation loops? Specifically, does the robot need to be embodied as part of the self to be effective, or does it function merely as a comforting tool? We aim to investigate this ambiguity (RQ1 & RQ2) by exploring whether a robotic arm can preserve the stress-reducing benefits of biological self-touch and how users conceptually categorize this interaction.

2.3 Supernumerary/Assistive Robotic Arms and Affective Touch

Supernumerary robotic limbs and assistive arms demonstrate that extra limbs can share workloads, stabilize posture, and free biological hands.[2, 8, 17, 20, 23] Research on devices like the Third Thumb and virtual appended limbs reveals rapid learnability and neural reorganization, highlighting the feasibility of embodied augmentation for functional tasks [10, 24, 26]. Similarly, wheelchair-mounted arms (e.g., JACO) significantly enhance independence [1, 2].

However, these systems primarily target physical manipulation. In the affective domain, robot-initiated touch (e.g., HuggieBot) utilizes softness and warmth to improve comfort [4]. Yet, research

examining interactions—where a robotic arm acts as a bodily extension for self-regulation rather than task execution or passive reception—remains scarce. It is unclear if a robot performing a "self-touch" gesture provides similar cognitive or emotional benefits to biological self-touch, or if it is merely perceived as an external tool.

3 Method

3.1 Apparatus

See Figure 1 for the system setup. We used an Agilex 6-DoF robotic arm.¹ mounted chair-side to simulate a supernumerary limb. An open-source 3D-printed prosthetic hand served as the end-effector in a fixed chin-resting pose. Paxini pressure sensors (resolution 0.01 N)² were embedded on the back of the hand to record contact forces during both self-touch and robotic touch conditions. A PC running Ubuntu 22.04 controlled the system using a "teach-and-repeat" trajectory for standardized motion replay. A physical emergency stop button was provided. Physiological signals (BVP, EDA, respiration) were recorded using a PhysioLAB Tech wearable sensor suite.³

During robotic conditions, participants received multimodal feedback: tactile feedback from the prosthetic hand contacting their face, visual feedback from the arm moving into position in their peripheral view, and auditory cues from the motor noise accompanying arm motion. Participants can see the arm throughout its trajectory and felt the contact force once the hand reached the chin-support pose. mk9

3.2 Participants

We recruited six participants (all male; age $M = 27.3$, $SD = 2.2$ years). All were right-handed and reported no neurological or tactile impairments. Participants provided written informed consent in accordance with institutional ethical guidelines. The study protocol was approved by the HKUST(GZ) Institutional Review Board (IRB). The study followed a within-subjects design; each session lasted about 30 minutes (sensor calibration, blocks, questionnaires, and interview). Participants volunteered without monetary compensation. As an early-stage exploratory study aimed at assessing feasibility and refining the protocol, we used a small convenience sample.

3.3 Procedure

After a briefing and calibration block (5 arithmetic problems, no time limit, no word memory), the system recorded each participant's mean response time and set the initial time limit for formal trials at 1.2× the mean (clamped to 2–10 s). Participants then worked with the experimenter to find a comfortable robot chin-support angle and target position and adjusted how close the arm should approach. They completed four conditions in a Latin-square counterbalanced order: (A) Bare Hand, (B) No Gesture, (C) Naked Robot, and (D) Clothed Robot.

Cognitive task. We employed a dual-task paradigm combining an adaptive MIST variant [12] with an interleaved word-recall task. Each condition comprised 10 trials. On every trial, a three-operand arithmetic expression (e.g., $23 + 15 - 8$) appeared with five multiple-choice options (one correct, four near-miss distractors) and

a color-coded countdown bar (green→orange→red). The system tracked accuracy over the most recent 12 trials and dynamically adjusted the time limit and operand range to maintain ≈50% accuracy, ensuring sustained stress induction; timeouts were scored as errors. After each arithmetic response, a bilingual word (e.g., "Apple") was displayed for 2 s; participants memorized it while preparing for the next problem. At the end of each block, a recognition test presented 20 candidates (10 targets + 10 distractors) in random order; participants selected those they recalled from the block. We recorded arithmetic accuracy, response time, timeout count, hit rate, and false-alarm rate. In (A), participants rested their hand against their cheek in a "thinking gesture" while working. In (B), they kept both hands on the keyboard, refraining from self-touch. In the robotic conditions (C/D), the arm moved into a chin-supporting pose; participants controlled its engagement via a button press. Condition (C) used a bare plastic hand, while (D) added a soft sleeve.

3.4 Data Analysis

During each block, we logged task performance and recorded physiology (BVP, EDA, respiration) and contact force. After each condition, participants completed the NASA-TLX; the AvatarEQ was administered only in the robot conditions (C/D) [21]. The session ended with a semi-structured interview about how the robot was perceived (e.g., tool vs. body part) and its impact on comfort.

Data were exported to CSV and analyzed offline in Python. From the recorded signals (BVP, EDA, and respiration), we computed key physiological metrics, including Heart Rate (HR) and Heart Rate Variability (RMSSD and LF/HF ratio) from BVP, and Electrodermal Activity features from EDA (Skin Conductance Level (SCL) and mean frequency). Given the small sample size ($N = 6$), we focus on descriptive trends and non-parametric effect sizes (Kendall's W) rather than underpowered significance tests. We used descriptive statistics and Friedman tests (Kendall's W) to assess consistency, with exploratory paired comparisons for Bare Hand vs. No Gesture, Naked vs. Clothed Robot, and HasArm vs. NoArm. Interview recordings were transcribed and analyzed using an inductive thematic analysis: two authors iteratively coded responses, discussed discrepancies, and consolidated codes into themes.

4 Results

4.1 Physiological Indicators

Figure 2 and Kendall's W show consistent patterns.

Validation of Self-Touch Effect (A vs. B): We examined the baseline effect of self-touch. Compared with No Gesture (B), Bare Hand (A) showed higher RMSSD and lower SCL, indicating a more relaxed state. This aligns with prior work on stress-buffering self-touch [14, 19], supporting our protocol's sensitivity.

Robotic Support Effects (C/D): In robot conditions, indicators suggested reduced stress-related arousal. We highlight those with the highest cross-participant consistency (Kendall's W):

- **EDA Mean Frequency ($W \approx 0.34$):** This frequency-domain metric of skin conductance showed the strongest consistency among physiological signals, outperforming amplitude-based measures (SCL). Trends indicated lower sympathetic arousal frequency during robot support.

¹<https://global.agilex.ai/products/piper>

²<https://mall.paxini.com/product/itpu/6894488c74e45ce616705b80>

³<https://physiolab.com/>

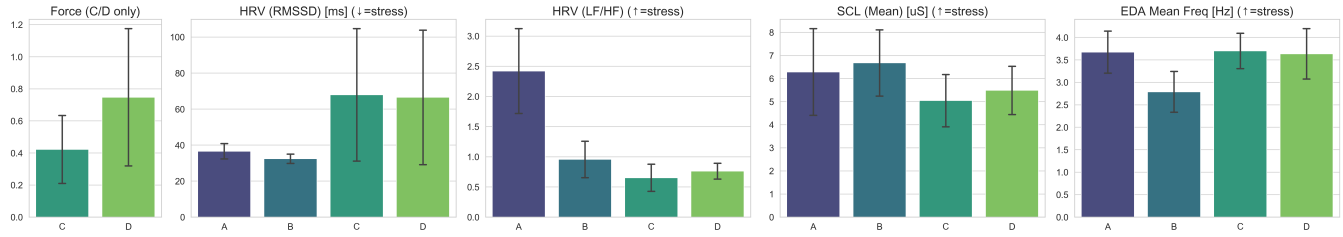


Figure 2: Physiological and mechanical indicators across conditions.

- **HRV LF/HF Ratio** ($W \approx 0.33$): As a marker of sympatho-vagal balance, LF/HF showed moderate consistency. Lower ratios in robot conditions suggest a shift away from sympathetic dominance.

Although statistical significance was not reached, the convergence of these indicators—especially EDA MeanFreq and HRV LF/HF—provides preliminary evidence that the robotic “third arm” can modulate physiological stress. Force data confirmed participants actively leaned on the robot, with slightly higher contact forces in the Clothed condition (D), suggesting the appearance invited more reliance.

Word-Recall Performance: Mean correct recognitions (out of 10) were similar across conditions: Bare Hand $M = 7.00$ ($SD = 2.00$), No Gesture $M = 6.83$ ($SD = 1.72$), Naked Robot $M = 6.00$ ($SD = 2.10$), and Clothed Robot $M = 7.17$ ($SD = 1.33$); Kendall’s $W \approx 0.15$. The introduction of the robotic arm did not noticeably impair secondary-task memory performance.

4.2 Subjective Workload (NASA-TLX) and Embodiment (AvatarEQ)

Subjective workload scores indicated visible differences between conditions. Kendall’s W revealed moderate effect sizes for both the Total Workload score ($W \approx 0.34$) and the Mental Demand subscale ($W \approx 0.30$). This suggests a perceptible shift in subjective load across conditions. However, unlike the physiological data, which hinted at relaxation, self-reported workload did not decrease; instead, there was a slight upward trend in reported demand during robot interactions, likely due to the attention required to coordinate with the device.

AvatarEQ varied little between robot conditions (C vs. D): effect sizes for Ownership and Appearance were negligible ($W \approx 0.0$), and other subscales were small ($W \leq 0.22$). The sleeve manipulation did not produce measurable differences in embodiment. Interviews echo this: the robot felt relaxing and stable, yet clearly non-embodied. Participants reported that the physical support translated to psychological relief. P1 noted, “*More like a bracket to lean on; it relieves some physical pressure; seems to reduce some agitation.*” P2 added that it acted as “*a place to lean on; makes the whole person feel more relaxed.*”

Participants also noted performance benefits but coordination overhead. P4 observed improved speed: “*With the robotic arm I had time to click; performance was indeed better; answered faster.*” P6 explained the added load: “*fitting myself to it becomes three tasks; workload actually increases a bit.*”

Finally, embodiment scores were reflected in strong descriptions of the arm as an external object. P1 described it as “*a foreign object, more like a head support bracket,*” while P4 stated clearly, “*I didn’t feel it was an extension of my body; it is just a robot arm.*” P6 echoed this, calling it “*more like a supporting shelf.*” Thus, the system functioned as an effective external aid rather than an integrated body part.

5 Discussion

Our exploratory study asked whether an external supernumerary robotic arm (a “third limb”) can reproduce the regulatory benefits of human self-touch during a demanding task. For RQ1 and RQ2, results suggest a clear pattern: robotic head support can attenuate stress-related physiological responses, even when users do not experience the robot as part of their own body.

Addressing RQ1, results point to a trade-off between physiological relief and cognitive demand. The robotic conditions tended to reduce indicators of sympathetic arousal (e.g., lower LF/HF and tonic EDA), consistent with prior self-touch findings [14, 19]. However, subjective workload did not decrease and sometimes slightly increased. This apparent contradiction with the title’s “stress reduction” framing reflects two distinct channels: physiological arousal (autonomic nervous system) versus perceived cognitive load. Unlike spontaneous self-touch, which occurs automatically and can offload cognitive resources [11], the robotic arm introduced coordination overhead—participants had to press a button to trigger the arm, monitor its approach in peripheral vision, and physically align themselves with a pre-taught trajectory while performing the dual task. As P6 noted, “*fitting myself to it becomes three tasks.*” This additional attentional cost likely drove the subjective workload increase even as the physical contact itself attenuated autonomic stress markers. Thus, the system may support bodily stress regulation while simultaneously imposing a cognitive coordination cost that future adaptive-control designs should aim to minimize.

Addressing RQ2, we examined whether the device must be “embodied” to be effective [10, 26]. Participants consistently rated the limb as an external tool (low ownership), yet still derived physiological relaxation and used it for support (leaning). This suggests that “friendliness” or “comfort” may be a more immediate design target than full “incorporation” for affective robotics. The sleeve manipulation did not produce measurable changes in embodiment scores ($W \approx 0.0$ for Ownership). A likely explanation is that participants were focused on the arithmetic and word-recall tasks, limiting their attentional resources for consciously evaluating the arm’s appearance; the brief, single-session exposure was also likely insufficient to shift body-schema integration. However, the sleeve may have operated at a less conscious level: the clothed condition consistently

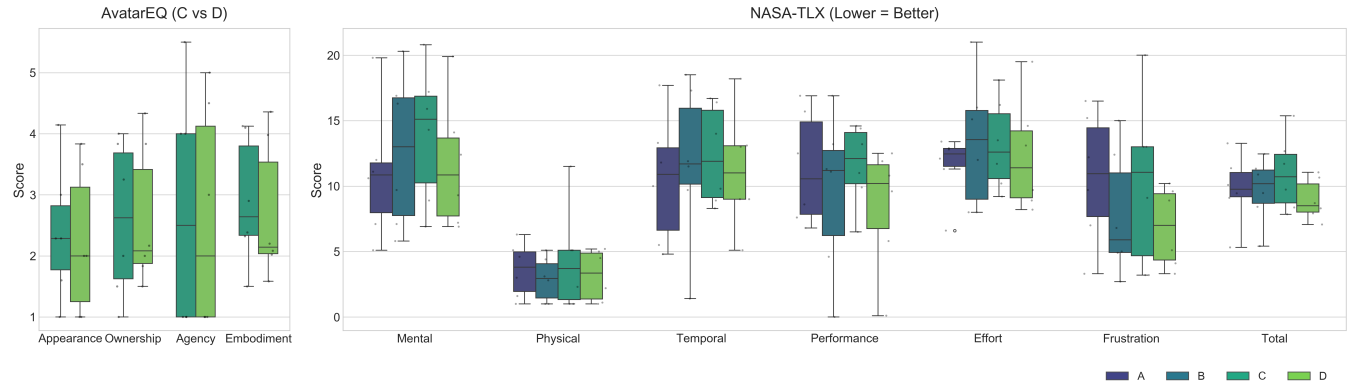


Figure 3: Avatar and NASA-TLX scores across conditions.

elicited higher leaning forces, suggesting that participants subconsciously trusted or felt more comfortable with the softer material upon contact. This aligns with hugging-robot findings that material and appearance cues shape acceptance [4], and extends them to user-initiated functional support.

Assistive robotic arms are typically framed as functional supports for people with disabilities, enabling independence in daily activities [2, 23]. Our findings suggest a complementary direction: these arms could also serve as everyday, user-controlled affective self-regulation tools—for example, providing desk-side chin support during demanding work for the general population.

This broader framing may reduce the social “markedness” of assistive devices, as symbolism and aesthetics profoundly shape social perception [13]. Normalizing non-clinical, everyday use helps lessen stigma, facilitating dignity-sensitive help-seeking [25]. Ultimately, shifting the perception of assistive robots towards universal design promotes social inclusion, enabling wider access to both functional and emotional support.

6 Limitations and Future Work

The primary limitation is the small and homogeneous sample ($N = 6$, all male, convenience recruitment). This constrains statistical power and generalizability; all physiological and behavioral trends reported here should be regarded as tentative until replicated with larger, gender-balanced cohorts. Individual differences in stress reactivity, tactile sensitivity, and movement artifacts may have disproportionately influenced the small-sample distributions.

Although we opted for a chair-side design to reduce physical burden compared to wearable systems and employed explicit button-triggered control to enhance user agency, future work could explore wearable supernumerary limbs to test whether closer body integration fosters stronger limb ownership. Furthermore, echoing participants’ feedback, future systems should close the loop with camera-based (vision) user feedback and automatically adjust the arm pose/trajectory to accommodate the user’s posture and target location, instead of requiring users to fit themselves to a pre-taught trajectory—this adaptation could also reduce the coordination overhead that elevated subjective workload in the current design. Finally, longitudinal experiments are necessary to verify whether such external tools can eventually drift toward the

“self” pole of the embodiment spectrum as suggested in tool-use literature [18].

7 Conclusion

Self-touch and “thinking gestures” are natural strategies people use to regulate stress and support cognition. As robotic augmentation moves toward closer body–robot coupling, a key question arises: can robot-mediated contact preserve these regulatory benefits? This exploratory study investigated whether a supernumerary robotic arm acting as a “third limb” to perform a head-support gesture could replicate the stress-buffering effects of biological self-touch during a high-demand task. Due to the limited sample size, we focused on directional trends rather than statistical significance. Our results show a divergence: physiologically, robotic support tended to produce signals associated with relaxation (e.g., higher HRV, lower tonic EDA), yet participants did not report lower subjective workload, instead experiencing slightly higher mental demand, likely due to interaction overhead. Furthermore, while the “clothed” robot condition offered slight comfort benefits, neither condition successfully induced a strong sense of embodiment or ownership; participants consistently viewed the robot as an external tool. These findings suggest that robotic touch can provide physiological stress relief without being fully embodied as “self”.

8 Declaration of AI Usage

We used Nano Banana Pro to assist in grammatical editing and data processing; all authors reviewed and verified all LLM-assisted outputs. The concept sketch in Fig. 1 (right) was co-created with the help of an AI image generation model.

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